

Word budget and cut plan

Measured after the two cuts and the data-section work. `texcount` on every element.
53 pages, 0 errors, 0 undefined references.

1. Every element, counted

Counts against the limit			Excluded from the limit		
1	Introduction	254		Executive summary	378
2	Introductory notes	349	A	Ontology	2,845
3	Tokenisation	107	B	Bayesian specifications	3,301
4	Phase locking	1,043	C	Light TSMixup	793
5	Ontology domain	340	D	KernelSynth	353
6	Expected behaviour	1,741	E	Retraining	588
7	Signals and data	342	F	Exploratory sweep	3,219
8	Experimental framework	1,230			
9	Results	174		Appendix total	11,099
10	Mitigation	637			
11	Agentic deployment	393		Also excluded: every table, figure,	
12	Security	1,404		caption, listing and the bibliography.	
13	Conclusions	258			
	Body	8,272			
	Thirteen titles	+34			
	Total	8,306	Limit		8,800

The report is currently 494 **words under**. The problem is what is still owed:

§8 Results, 174 → 600	+426	the working target
§8.3.2 Bayesian-analysis opening, replacing the TODO	+50	
Methodology scope paragraph	+62	written, ready to paste
§10.3 and §12.2 red placeholders → posteriors	≈ 0	they already occupy 296 words
Projected finish	8,844	
Against 8,800	-44	

2. Where the words are, inside the body

Only four sections are large enough to cut from without damage. Their internal breakdown:

§6 Expected behaviour 1,741		§8 Experimental framework 1,230	
lead-in	164	lead-in	48
H1 / H2 / H3 statements	239	Patch geometries	91
H1	350	Retraining	123
H2	208	Analysis design	100
H3	453	Exploratory frequency sweep	410
Mitigation claim	153	Bayesian analysis	448
Decisions and validation	135		
§12 Security 1,404		§4 Phase locking 1,043	
lead-in	165	lead-in (derivation)	495
Risk Management Framework	292	Redundancies and invariances	85
Adversarial-masking threat model	304	the three geometry cases	183
Threat models beyond it	227	Representation-level proposition	255
Controls, residual, conclusion	394		

3. The cut plan, in priority order

Every cut below removes text that is **said in full somewhere else in the document**. None removes an argument, and none touches a lecturer comment. You need 44 words; the list offers 470, so stop as soon as you are comfortable.

#	Target	now	save	Why it is safe
1	§12.4 closing paragraph	394	80	Its final third restates what §13 Conclusions now says — geometry as a deployment parameter, monitoring upstream. That duplication is new, created by adding the Conclusions section. Cut the closing, keep the residual-exposure argument.
2	§8.3.1 Exploratory frequency sweep	410	90	Appendix F carries this in 3,219 words and is about to carry more. The body needs what the sweep is, what it is not, and the pointer — not the method.
3	§8.3.2 Bayesian analysis	448	100	The model-by-model walk-through duplicates Table 2 and Appendix B. Keep the fail-closed paragraph verbatim: it answers comment C12. Cut the sentence enumerating what A, B, C, D1 and D2 each do.
4	§6.3 H3	453	80	The longest hypothesis subsection, and Appendix B specifies D1 and D2 in full. Keep both branch statements and the identification asymmetry; cut the restatement of the fitting procedure.
5	§6 hypothesis framing	403	60	H1, H2 and H3 are each stated twice: once as a subsubsection* near the top, once as a numbered subsection below. Merge, or cut the first statement to one sentence each.
6	§10.1 Further strategies	294	60	Requirement 8 asks for <i>at least one</i> mitigation with a principled explanation, and the overlap claim delivers that on its own. The second and third strategies are descriptive and can be stated more briefly.
Available		470	Needed:	44

How to do it

Cut 1 alone closes the gap with 36 to spare. Taking 1 and 2 gives 126 of slack, which is what I would aim for, because §9 Results is the one section whose length you cannot yet predict — if the fits produce more than expected it will want 700 rather than 600, and that is exactly the situation the slack exists for.

Do them in order and re-count after each. `texcount -brief -sum` on the thirteen body files plus the 34 title words is the number that matters; the appendices, tables, captions and bibliography never enter it.

4. What not to cut

- **§4's derivation lead-in** (495 words). It is requirements 1 and 2 and it is not duplicated anywhere — Appendix B specifies the *models*, not the operator.
- **§12.1 Risk Management Framework** (292). It is requirement 10, and the four RMF functions appear nowhere else in prose.
- **§12.2 Adversarial-masking threat model** (304). It carries the five elements of lecturer comment B4 — capabilities, surface, telemetry, consequence, residual risk.
- **§4's representation-level proposition** (255). It is comment B5, in his own wording.
- **The scope paragraph** (62, not yet pasted). It answers three separate comments for less than any of them would cost individually.
- **§11 Agentic deployment** (393). Already cut from 679 this week; requirement 9 rests on it entirely.